

Server-built PDFs from phone photos

PoC analysis after testing: measured quality loss, risks for sensitive and multilingual documents, why threshold tuning is slow, and why a user-supplied PDF is the better architecture

Date: 2026-10-08 · Suite: 36 passed, 1 skipped (GEN_BACKEND=sync, tesseract 5.5.0) · Pipeline: load → detect → crop/deskew → orientation → align → enhance → compose

1. Finding — the PDF images are lower quality than the uploads

Each photo passes a chain where every step discards detail. Measured on two sample documents through `pipeline.process_image`:

sample	uploaded	in PDF	eff. DPI on A4	path / time
clean_scan	1200x1600 (1.92MP, 149KB)	1200x1600 (1.92MP, 142KB)	137 -> 137	trim fallback, 164ms total
photo_dark_bg	1700x2100 (3.57MP, 539KB)	1234x1628 (2.01MP, 339KB)	180 -> 139	poly warp, 99ms total

- **2000px cap (~171 DPI max).** Longest side is clamped to 2000px — a 12MP phone photo is cut to a quarter of its pixels before the PDF is built. Print/archival standard is 300 DPI.
- **Double JPEG compression.** The upload is already JPEG; the pipeline decodes, processes, and re-encodes at q82. Text edges gain ringing artifacts, compounded on every re-run.
- **CLAHE always on.** Local contrast (clipLimit 2.0) evens lighting but amplifies sensor noise and can wash out light stamps, pencil marks, and colored seals. No per-image decision.
- **Three resampling passes.** Perspective warp → deskew rotate → final resize (cubic-class filters, but triple interpolation still softens fine print, MRZ lines, seal micro-text).

Verdict: output is a *screen-readable facsimile*, not a faithful archival copy.

2. Finding — risks for sensitive and multilingual documents

risk	impact	status in this PoC
ID docs on server disk (raw + cache + derivatives + logs), never auto-purged	Breach exposure; retention liability	No expiry, no access log, no purge
Silent alteration (edge stamps clipped, faint signatures bleached) becomes the record	Evidentiary integrity	2% padding heuristic only; no diff vs raw
OSD orientation is Latin-tuned; Devanagari orients worse, lower confidence	Sideways Nepali docs abstain (2.0 gate) or mis-rotate	Needs tesseract-ocr-nep; untested
Mixed-script pages (English + Nepali, e.g. citizenship) worst for OSD + OCR	Garbage text layer, wrong rotation	No per-language quality signal
WARN tells you <i>something</i> happened, not whether text stayed legible	False confidence in bad output	By design; needs human review step

3. Finding — why threshold tuning on real photos is slow

Guide values (area 10–97%, rectangularity ≥ 0.90 , blur ≥ 100 , tilt $\geq 0.5^\circ$, OSD ≥ 2.0) were measured on synthetic images: clean fonts, even light, solid backgrounds. Real photos break each assumption, and the thresholds

interact:

- Otsu assumes a two-hump brightness histogram — wood grain, shadow, glare, off-white tables make it one-hump, so the 'paper shape' merges with the table or fractures.
- Blur ≥ 100 shifts with camera shake, noise, and content (guide's own readings: sharp 1121, light blur 211, medium 32, heavy 3) — one number cannot serve all phones.
- Rectangularity ≥ 0.90 rejects curled pages, folded corners, and reflective laminates — common for IDs.
- Calibration is a multi-dimensional search: 100+ real photos, per-image metric logs, eyeballing every rejection — weeks of labeled-data work, and thresholds rot as cameras and habits change.

4. Recommendation — ask the user for a finished PDF

- **Zero quality loss:** uploaded bytes are archived bytes — no DPI cap, re-encode, or CLAHE washout.
- **Zero tuning burden:** layout and legibility are verified by a human eye at upload (preview + 'is this readable?' check beats any Laplacian threshold).
- **Smaller blast radius:** one file instead of raw + cache + derivatives; retention/deletion trivial.
- **Language-independent:** Devanagari, mixed-script, handwritten — all pass through untouched.
- **No silent alteration:** what the applicant signed is what is stored.

Suggested shape: PDF upload as the primary path — validate (page count, size, text layer, first-page thumbnail + blur check), keep photo→PDF only as an assisted fallback ('no scanner? photograph each page; quality not guaranteed').

Appendix — measured stage timings (ms) and reference thresholds

stage	ms range
load	~10-15
detect	~18-25
crop/deskew	~15-30
orientation	~0.1
align	~20-25
enhance (CLAHE)	~15-80
compose	~4-5

Thresholds: warp needs area 10–97% + rectangularity ≥ 0.90 + 2nd contour $\leq 40\%$ (edge retry Canny 20/60 on failure); deskew at tilt $\geq 0.5^\circ$, limit 45° ; OSD accepted at confidence ≥ 2.0 ; blur ≥ 100 on 1000px copy; compose longest side 2000px, JPEG q82.